

# Linear correlation and Linear regression

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# Correlation coefficient

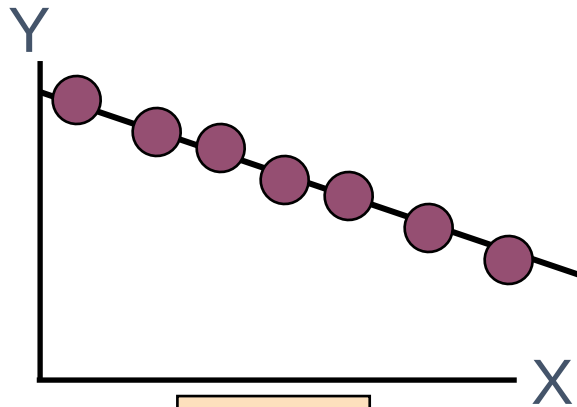
- Pearson's Correlation Coefficient is standardized covariance (unitless):

$$r = \frac{\text{covariance}(x, y)}{\sqrt{\text{var } x} \sqrt{\text{var } y}}$$

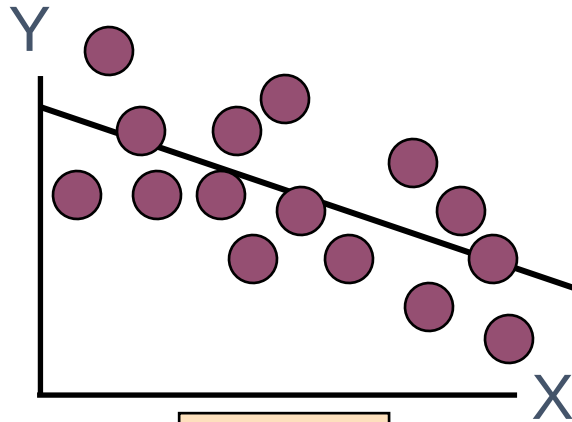
# Correlation

- Measures the relative strength of the *linear* relationship between two variables
- Unit-less
- Ranges between  $-1$  and  $1$
- The closer to  $-1$ , the stronger the negative linear relationship
- The closer to  $1$ , the stronger the positive linear relationship
- The closer to  $0$ , the weaker any positive linear relationship

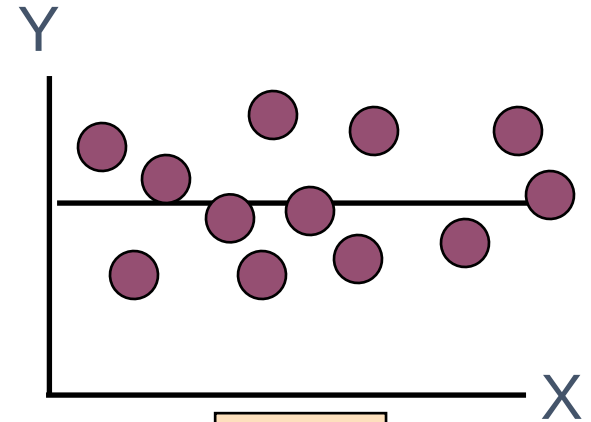
# Scatter Plots of Data with Various Correlation Coefficients



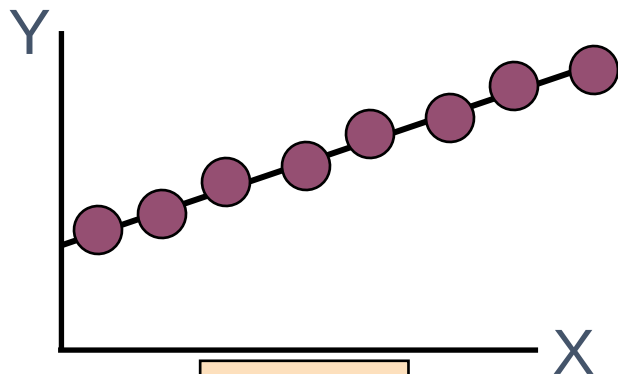
$$r = -1$$



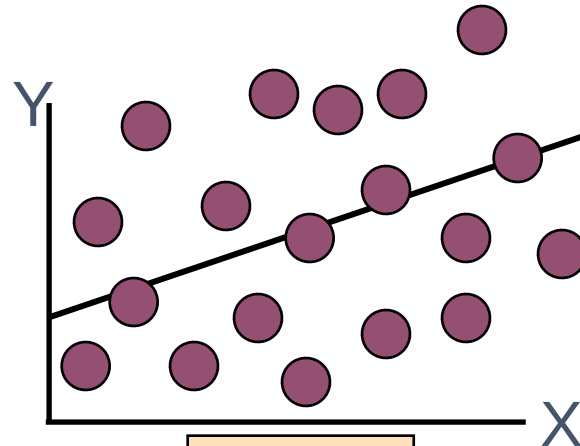
$$r = -.6$$



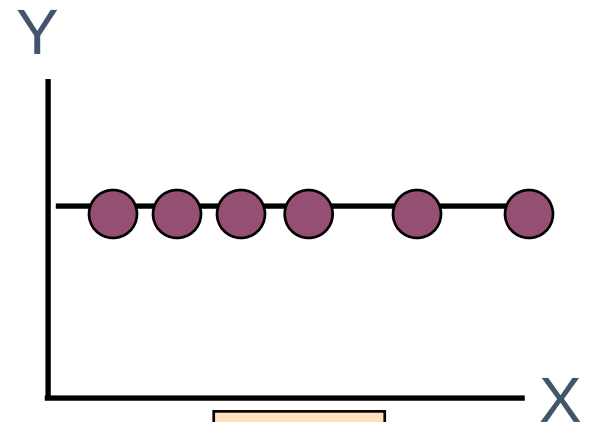
$$r = 0$$



$$r = +1$$



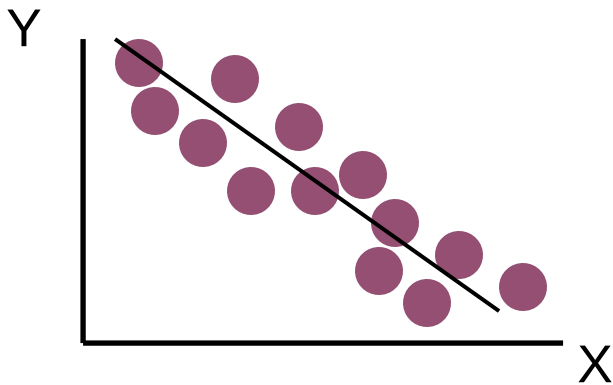
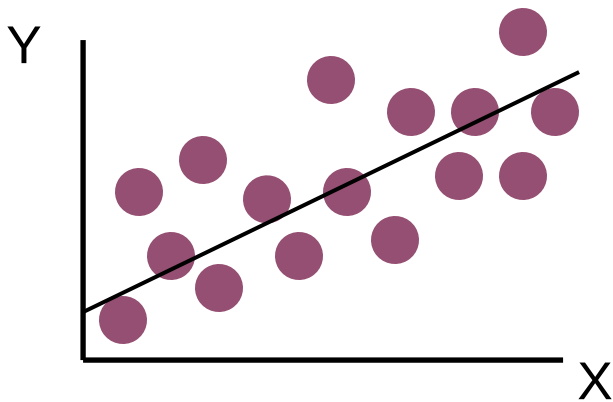
$$r = +.3$$



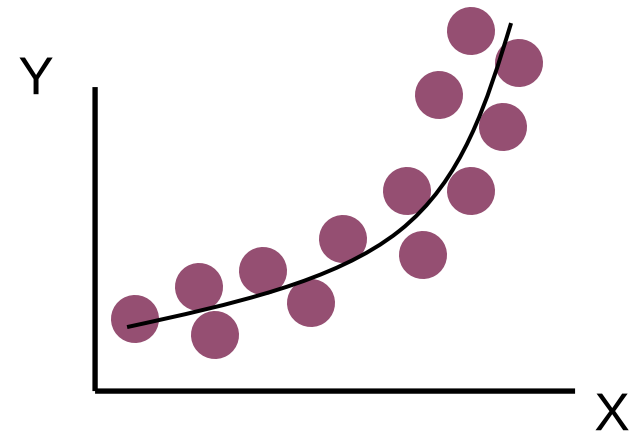
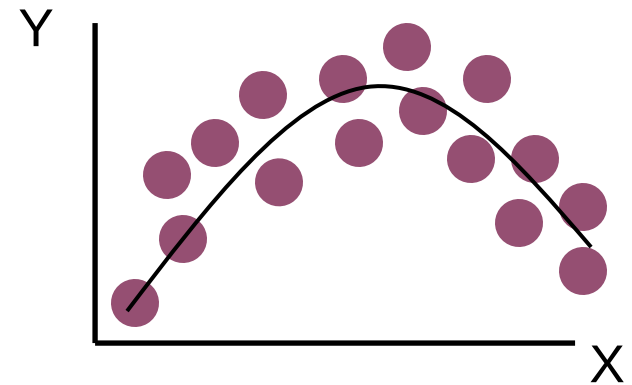
$$r = 0$$

# Linear Correlation

Linear relationships

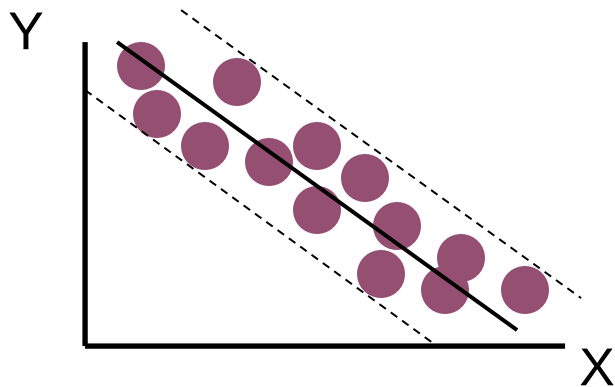
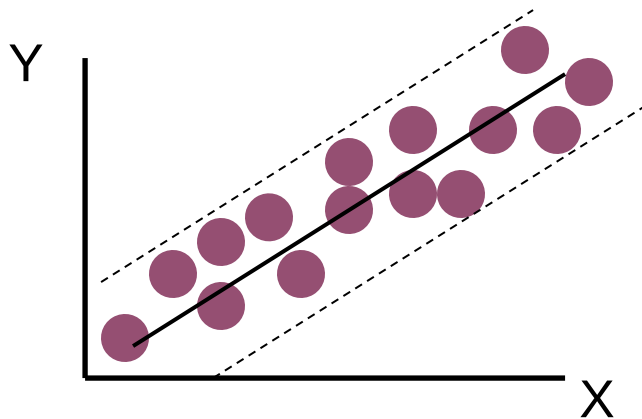


Curvilinear relationships

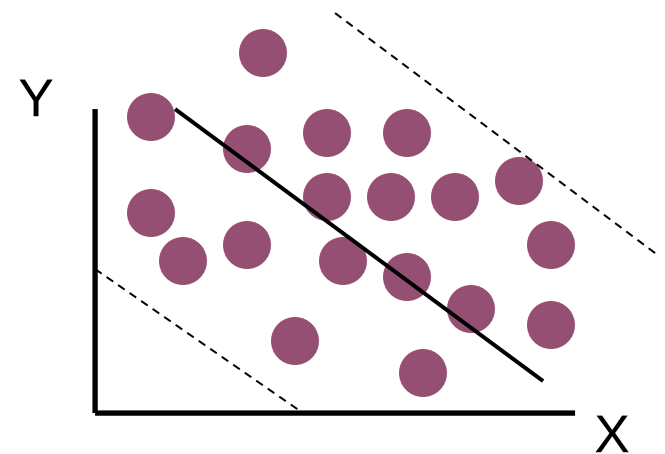
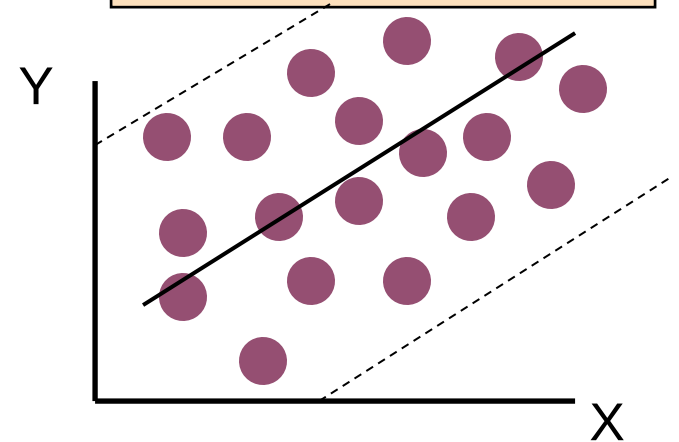


# Linear Correlation

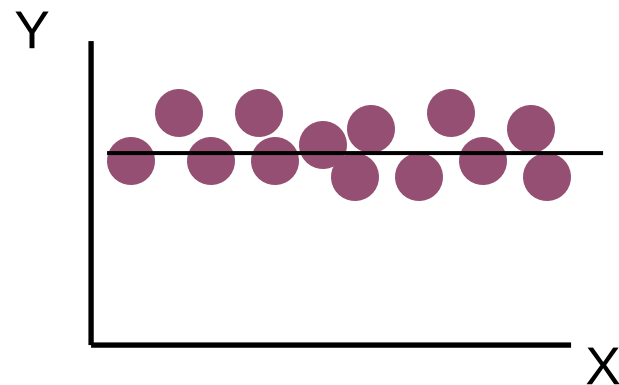
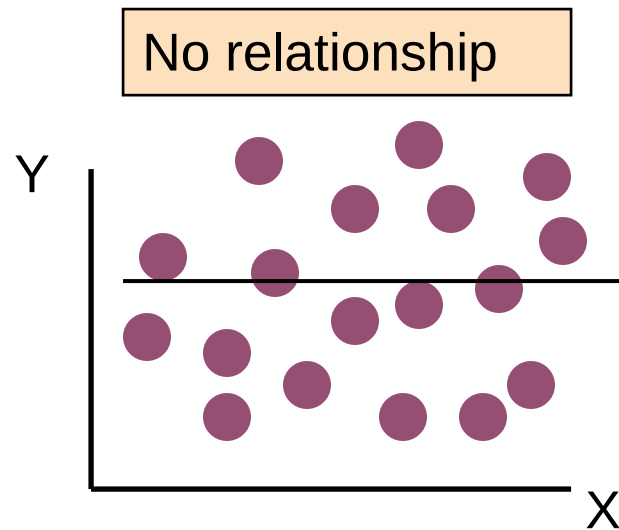
Strong relationships



Weak relationships



# Linear Correlation



Simple calculation formula...

### **Formula for the Correlation Coefficient $r$**

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n(\sum x^2) - (\sum x)^2][n(\sum y^2) - (\sum y)^2]}}$$

where  $n$  is the number of data pairs.

# Example

## Car Rental Companies

Compute the correlation coefficient for the data

From car rental companies

### Solution

**Step 1** Make a table as shown here.

| Company | Cars $x$<br>(in ten thousands) | Income $y$<br>(in billions) | $xy$ | $x^2$ | $y^2$ |
|---------|--------------------------------|-----------------------------|------|-------|-------|
| A       | 63.0                           | 7.0                         |      |       |       |
| B       | 29.0                           | 3.9                         |      |       |       |
| C       | 20.8                           | 2.1                         |      |       |       |
| D       | 19.1                           | 2.8                         |      |       |       |
| E       | 13.4                           | 1.4                         |      |       |       |
| F       | 8.5                            | 1.5                         |      |       |       |

# Example

**Step 2** Find the values of  $xy$ ,  $x^2$ , and  $y^2$  and place these values in the corresponding columns of the table.

The completed table is shown.

| Company | Cars $x$<br>(in 10,000s) | Income $y$<br>(in billions) | $xy$                 | $x^2$                  | $y^2$                |
|---------|--------------------------|-----------------------------|----------------------|------------------------|----------------------|
| A       | 63.0                     | 7.0                         | 441.00               | 3969.00                | 49.00                |
| B       | 29.0                     | 3.9                         | 113.10               | 841.00                 | 15.21                |
| C       | 20.8                     | 2.1                         | 43.68                | 432.64                 | 4.41                 |
| D       | 19.1                     | 2.8                         | 53.48                | 364.81                 | 7.84                 |
| E       | 13.4                     | 1.4                         | 18.76                | 179.56                 | 1.96                 |
| F       | 8.5                      | 1.5                         | 2.75                 | 72.25                  | 2.25                 |
|         | $\Sigma x = 153.8$       | $\Sigma y = 18.7$           | $\Sigma xy = 682.77$ | $\Sigma x^2 = 5859.26$ | $\Sigma y^2 = 80.67$ |

# Example

**Step 3** Substitute in the formula and solve for  $r$ :

$$\begin{aligned} r &= \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n(\sum x^2) - (\sum x)^2][n(\sum y^2) - (\sum y)^2]}} \\ &= \frac{(6)(682.77) - (153.8)(18.7)}{\sqrt{[(6)(5859.26) - (153.8)^2][(6)(80.67) - (18.7)^2]}} = 0.982 \end{aligned}$$

The correlation coefficient suggests a strong relationship between the number of cars a rental agency has and its annual income.

# Simple Regression

- Simple regression analysis is a statistical tool That gives us the ability to estimate the mathematical relationship between a dependent variable (usually called  $y$ ) and an independent variable (usually called  $x$ ).
- The dependent variable is the variable for which we want to make a prediction.

# Introduction

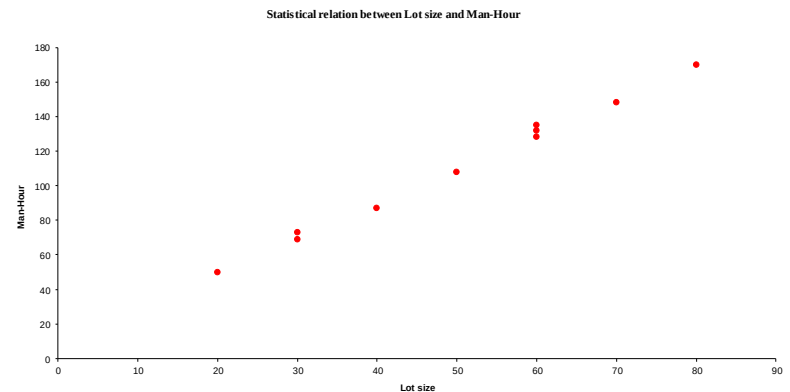
- The primary goal of quantitative analysis is to use current information about a phenomenon to predict its future behavior.
- Current information is usually in the form of a set of data.
- In a simple case, when the data form a set of pairs of numbers, we may interpret them as representing the observed values of an independent (or predictor ) variable  $X$  and a dependent ( or response) variable  $Y$ .

| lot size | Man-hours |
|----------|-----------|
| 30       | 73        |
| 20       | 50        |
| 60       | 128       |
| 80       | 170       |
| 40       | 87        |
| 50       | 108       |
| 60       | 135       |
| 30       | 69        |
| 70       | 148       |
| 60       | 132       |

# Introduction

- The goal of the analyst who studies the data is to find a functional relation

$y = f(x)$   
between the response  
variable  $y$  and the predictor  
variable  $x$ .



# General regression model

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

1.  $\beta_0$ , and  $\beta_1$  are parameters
2.  $X$  is a known constant
3. Deviations  $\varepsilon$  are independent  $N(0, \sigma^2)$

# Meaning of Regression Coefficients

- The values of the regression parameters  $\beta_0$ , and  $\beta_1$  are not known. We estimate them from data.
- $\beta_1$  indicates the change in the mean response per unit increase in  $X$ .
- $\beta_0$  indicates the average of  $y$  when  $x=0$ .

# Regression Line

- Using calculus, we obtain estimating formulas:

$$\hat{\beta}_1 = b_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{n \sum_{i=1}^n x_i y_i - \sum_{i=1}^n x_i \sum_{i=1}^n y_i}{n \sum_{i=1}^n x_i^2 - (\sum_{i=1}^n x_i)^2}$$

$$\hat{\beta}_0 = b_0 = \bar{y} - b_1 \bar{x}$$

# Example

- The weekly advertising expenditure (x) and weekly sales (y) are presented in the following table.

| y    | x  |
|------|----|
| 1250 | 41 |
| 1380 | 54 |
| 1425 | 63 |
| 1425 | 54 |
| 1450 | 48 |
| 1300 | 46 |
| 1400 | 62 |
| 1510 | 61 |
| 1575 | 64 |
| 1650 | 71 |

$$n = 10 \quad \sum x = 564 \quad \sum x^2 = 32604$$

$$\sum y = 14365 \quad \sum xy = 818755$$

- The estimates of the regression coefficients are:

$$b_1 = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2} = \frac{10(818755) - (564)(14365)}{10(32604) - (564)^2} = 10.8$$

$$b_0 = 1436.5 - 10.8(56.4) = 828$$

# Point Estimation of Mean Response

- The estimated regression function is:

$$\hat{y} = 828 + 10.8x$$

$$\text{Sales} = 828 + 10.8 \text{ Expenditure}$$

- This means that if the weekly advertising expenditure is increased by \$1 we would expect the weekly sales to increase by \$10.8.

- Fitted values for the sample data are obtained by substituting the x value into the estimated regression function.
- For example if the advertising expenditure is \$50, then the estimated Sales is:

$$\text{Sales} = 828 + 10.8(50) = 1368$$

- This is called the point estimate (forecast) of the mean response (sales).

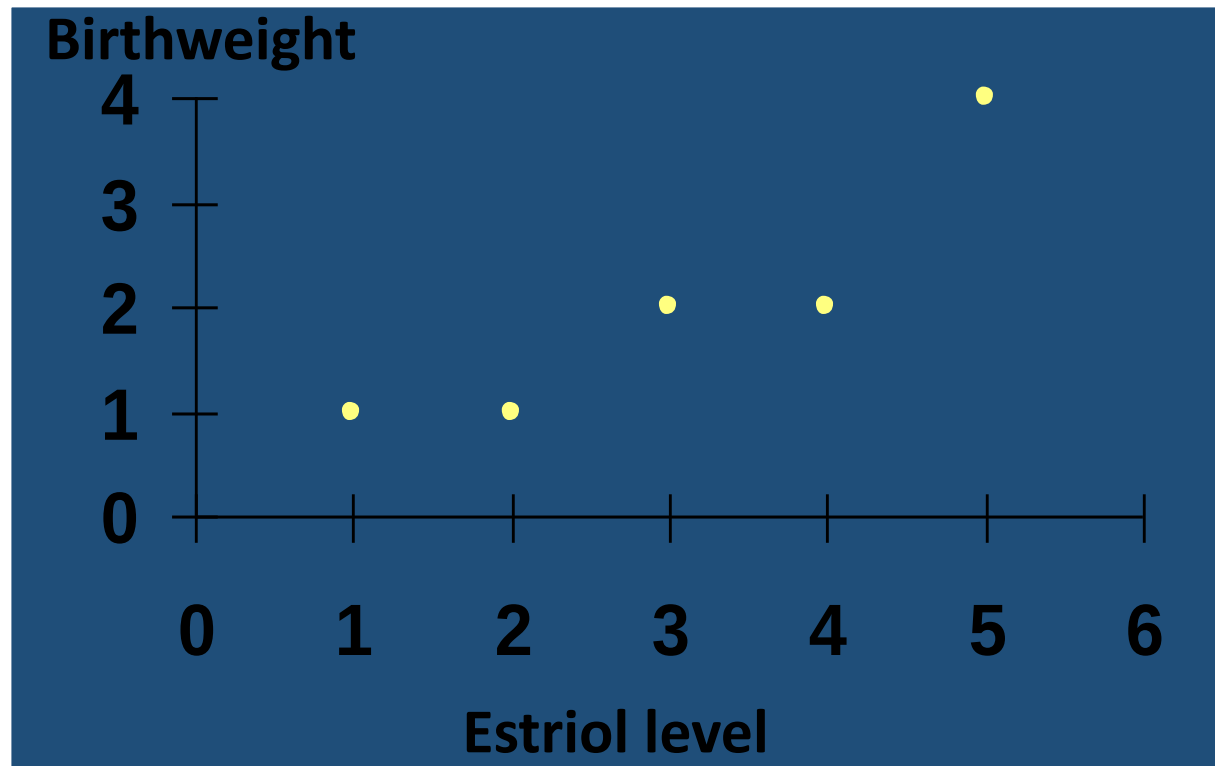
# Parameter Estimation Example

- What is the **relationship** between Mother's Estriol level & Birthweight using the following data?

| <u>Estriol</u> | <u>Birthweight</u> |
|----------------|--------------------|
| (mg/24h)       | (g/1000)           |
| 1              | 1                  |
| 2              | 1                  |
| 3              | 2                  |
| 4              | 2                  |
| 5              | 4                  |

# Scatterplot

## Birthweight vs. Estriol level



## Parameter Estimation Solution Table

| $X_i$ | $Y_i$ | $X_i^2$ | $Y_i^2$ | $X_i Y_i$ |
|-------|-------|---------|---------|-----------|
| 1     | 1     | 1       | 1       | 1         |
| 2     | 1     | 4       | 1       | 2         |
| 3     | 2     | 9       | 4       | 6         |
| 4     | 2     | 16      | 4       | 8         |
| 5     | 4     | 25      | 16      | 20        |
| 15    | 10    | 55      | 26      | 37        |

## Parameter Estimation Solution

$$\hat{\beta}_1 = \frac{n \sum_{i=1}^n X_i Y_i - \left( \sum_{i=1}^n X_i \right) \left( \sum_{i=1}^n Y_i \right)}{n \sum_{i=1}^n X_i^2 - \left( \sum_{i=1}^n X_i \right)^2} = \frac{5 * 37 - (15)(10)}{5 * 55 - (15)^2} = 0.70$$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X} = 2 - (0.70)(3) = -0.10$$